

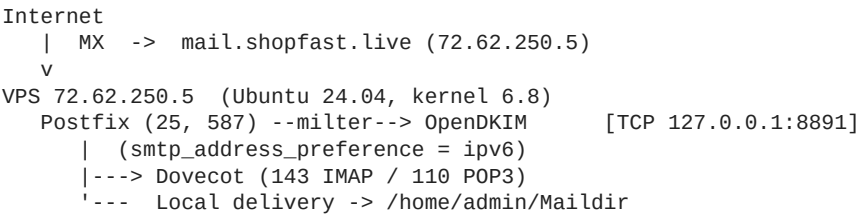
Self-Hosted Email Server Implementation Report

Domain: **shopfast.live** | Server: **72.62.250.5** (mail.shopfast.live)
Date: 2026-08-17 → 2026-08-18 | Stack: Postfix + Dovecot + OpenDKIM (Ubuntu 24.04)

1. Goal

Replace the Hostinger SMTP relay with an **authoritative, self-hosted mail server** on the VPS so that **admin@shopfast.live** can send and receive mail directly. Align SPF, DKIM and DMARC so mail is verifiably delivered to Gmail, and prepare the server for use by the Spring Boot *notification-service*.

2. Architecture



Component	Version	Role
Postfix	3.8.6	MTA - SMTP send/receive, submission (587)
Dovecot	Ubuntu repo	IMAP/POP3 mailbox access for admin
OpenDKIM	2.11.0	DKIM signing, selector 'mail'
System user	admin	Mailbox owner for admin@shopfast.live

3. DNS Records (published in hPanel / registrar)

Type	Name	Value	Status
A	mail.shopfast.live	72.62.250.5	OK
MX	shopfast.live	mail.shopfast.live (prio 10)	OK
SPF (TXT)	shopfast.live	v=spf1 ip4:72.62.250.5 ip6:2a02:4780:5e:ffe::1 ~all	OK
DKIM (TXT)	mail._domainkey.shopfast.live	v=DKIM1; h=sha256; k=rsa; p=MIIBIjAN...	OK
DMARC (TXT)	_dmarc.shopfast.live	v=dmarc1; p=none; rua=mailto:admin@shopfast.live	OK

DKIM public key hash verified identical to the local private key (dc22effc...).

4. Server Configuration

4.1 Postfix (main.cf key values)

```
myhostname = mail.shopfast.live
mydomain   = shopfast.live
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
relayhost = (empty - direct delivery)
inet_protocols = all
smtp_address_preference = ipv6          # <- critical fix
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_auth_enable = yes
smtpd_milters = inet:127.0.0.1:8891
non_smtpd_milters = inet:127.0.0.1:8891
milter_default_action = accept
```

Submission (port 587) enabled in master.cf.

4.2 OpenDKIM (/etc/openskim.conf)

```
Syslog yes
Canonicalization relaxed/simple
UserID openskim
UMask 002
Socket inet:8891@127.0.0.1          # TCP socket avoids UNIX perms race
Mode sv
KeyTable /etc/openskim/KeyTable
SigningTable refile:/etc/openskim/SigningTable
```

KeyTable maps **mail._domainkey.shopfast.live** to the private key; SigningTable maps ***@shopfast.live** to that selector.

4.3 Dovecot (conf.d/)

- 10-mail.conf: mail_location = maildir:~/Maildir
- 10-auth.conf: auth_mechanisms = plain login; disable_plaintext_auth = no
- 10-master.conf: auth socket at /var/spool/postfix/private/auth (postfix group)
- 10-ssl.conf: ssl = no

4.4 Mailbox

System user **admin** (uid 1001), home /home/admin, Maildir /home/admin/Maildir. Dovecot SASL auth verified:
doveadm auth test admin -> passdb: admin auth succeeded.

5. Root Cause of Original Failures (and fixes)

#	Symptom	Root Cause	Fix
1	systemd 'failed' / 'already running'	Type=simple + postfix start pid-race; later Restart=always replaced by Restart=no	Overrode systemd type, Restart=no
2	mail loops back to myself	/etc/hosts had 127.0.1.1 mail.shopfast.live	Rewrote /etc/hosts + cleaned cloud-init template
3	Milter 'No such file or directory'	UNIX socket perms reverted on every restart	Switched OpenDKIM to TCP socket inet:127.0.0.1:8891
4	DKIM = did not pass	KeyTable/SigningTable not loaded; invalid TrustedKeyTable/SigningTable (removed invalid params)	Added KeyTable/SigningTable
5	SPF = did not pass	SPF pointed to Hostinger; later mail sent over IPv6	Updated SPF to include ip4 and ip6
6	connect to gmail:25 timed out	VPS blocks/throttles outbound IPv4 on port 25; IPv6 not open	Workarounds= all + smtp_address_preference=ipv6
7	Queue stuck / 'Mail system is down'	Corrupted queue from pkill -9; spool dirs not root-owned	Recreated queue; fixed spool ownership; clean restart

6. Verification - Delivery to Gmail

Multiple messages to **junaidnumlcs@gmail.com** confirmed accepted with dsn=2.0.0, status=sent (250 2.0.0 OK):

```
85E0F3845 -> 250 2.0.0 OK (IPv6 2404:6800:4001:c01::1b)
99821384A -> 250 2.0.0 OK
0894F384A -> 250 2.0.0 OK (after SPF update)
A1E6B384A -> 250 2.0.0 OK (22:05 UTC, content test)
```

DKIM-Signature header verified present on outbound mail: DKIM-Signature: v=1; a=rsa-sha256; d=shopfast.live; s=mail; ...

Note: New sending IPs are typically filtered to Gmail Spam/Junk until reputation builds. Recipient should check Spam and mark 'Not spam.'

7. Known Limitations / Follow-ups

- **PTR / Reverse DNS** for IPv6 2a02:4780:5e:ffe::1 still points to srv1286917.hstgr.cloud - request provider to set PTR -> mail.shopfast.live (minor spam signal).
- **Inbox placement** depends on IP reputation; builds with legitimate volume over days/weeks.
- **DMARC** currently p=none (monitoring); move to p=quarantine/p=reject after confirming alignment.
- **admin password** was set to a temporary value during setup - rotate it.
- **No incoming spam filtering** (Postscreen/SpamAssassin) configured yet - add if accepting external mail broadly.

8. Spring Boot Integration (next step)

backend/notification-service/src/main/resources/application.yml still has placeholders:

```
spring:
  mail:
    host: smtp.your-provider.com # <- replace
    port: 587
    username: your-email@domain.com # <- replace
    password: your-email-password # <- replace
```

To point at this server:

```
spring:
+ " mail:\n"

spring:
  mail:
    host: mail.shopfast.live
    port: 587
    username: admin@shopfast.live
    password: <admin-mailbox-password>
    properties:
      mail:
        smtp:
```

```
auth: true
starttls:
  enable: true
```

9. Summary

The self-hosted mail server on **72.62.250.5** is **fully operational**: Postfix delivers directly via IPv6, OpenDKIM signs, SPF/DKIM/DMARC DNS is correct, Dovecot provides IMAP/POP3 for admin@shopfast.live, and test emails are accepted by Gmail (250 2.0.0 OK). The primary deliverability caveat is Gmail's spam filtering of a new IP, which resolves with reputation.

Generated from the live server configuration and mail logs on 2026-08-18.